Detection Algorithms

This supplementary section presents detection algorithm implementations for the cognitive attack detection methods defined in Part 1. Where sec:pseudocode (Section 2a) presents the six core defense mechanisms (Firewall, Sandbox, Trust, Tripwires, Consensus, Drift Detection), this supplement presents the *detection analytics pipeline* that evaluates their output—including ROC analysis, multi-detector fusion, online/batch detection architectures, and false positive mitigation.

ROC Analysis Algorithms

Algorithm 1: ROC Curve Construction

[1] Detector D, attack samples X_{attack}, benign samples X_{benign}, threshold count n ROC curve, AUC, optimal threshold τ^* Compute scores: $S_{\text{attack}} \leftarrow [D(x) : x \in X_{\text{attack}}]$ Compute scores: $S_{\text{benign}} \leftarrow [D(x) : x \in X_{\text{benign}}]$ Generate thresholds: $T \leftarrow \text{linspace}(\min(S), \max(S), n)$ each $\tau \in T$ $\text{TPR}[\tau] \leftarrow |S_{\text{attack}} > \tau| / |X_{\text{attack}}|$ $\text{FPR}[\tau] \leftarrow |S_{\text{benign}} > \tau| / |X_{\text{benign}}|$ $\text{AUC} \leftarrow \int \text{TPR} \, d(\text{FPR})$ Trapezoidal integration